INFOTACT SOLUTIONS PROJECT DOCUMENT – ADVANCED DATA SCIENCE & ML ENGINEERING (VOL. II)

Real-World Applications - Hands-on System Building

To continue expanding your advanced Data Science and Machine Learning portfolio, we will focus on highly specialized, cutting-edge AI domains that are rarely covered in standard courses: **Causal Machine Learning**, **Hyperspectral Computer Vision**, and **Multimodal Time-Series Forecasting (Transformers)**.

*Here are three more unique, highly advanced Data Science and ML projects formatted to the **Infotact standard.***

Project 1 - "TerraSpectra": Hyperspectral Crop Disease Forecasting

Domain: Precision Agriculture & Advanced Computer Vision

Problem Statement: *Traditional satellite crop monitoring uses standard RGB or simple infrared imagery. These models only detect crop diseases after the leaves turn yellow or brown, which is too late to save the harvest. Standard 2D Convolutional Neural Networks (CNNs) cannot process the massive depth of data required for early detection.*

Use Case: An agricultural analyst monitors a 1,000-acre farm via the **TerraSpectra** dashboard. The system ingests Hyperspectral satellite data (which captures 200+ bands of light, invisible to the human eye). The backend 3D-CNN + Vision Transformer model analyzes the subtle chemical changes in the plants' chlorophyll reflection. The dashboard highlights a specific 5-acre zone in red, predicting a fungal blight outbreak **three weeks** before any visible symptoms appear on the leaves, allowing for hyper-targeted, preventative pesticide application.

Key Modules:

- **Geospatial Data Pipeline (Python, Rasterio, GDAL):** Ingests and normalizes massive, multi-gigabyte hyperspectral data cubes (e.g., from NASA's Hyperion or ESA's Sentinel missions).
- **3D-CNN & ViT Hybrid Model (PyTorch):** A complex neural network utilizing 3D convolutions to capture spatial features and a Vision Transformer to process the deep spectral bands.
- **Inference API (FastAPI):** A high-throughput REST API capable of chunking and processing massive satellite rasters on GPU.
- **GIS Dashboard (React & Deck.gl):** A 3D map interface rendering the predicted disease heatmaps overlaid on satellite topography.

Electronic city Bengaluru
Karnataka, PN 560100
support@infotact.in
+91 9124936538

Week-wise Development Plan:

Week	ML Engineering & Data Pipeline (PyTorch, Rasterio)	Frontend & GIS (React, Deck.gl)
Week 1	**Data Preparation:** Use Rasterio to parse mock hyperspectral data cubes (HDF5/GeoTIFF formats). Perform dimensionality reduction (PCA) on the 200+ spectral bands.	**Map Scaffolding:** Build React app. Integrate Deck.gl and Mapbox to render the base geographical farm layers.
Week 2	**Model Architecture:** Build the Hybrid 3D-CNN in PyTorch. Train it to classify healthy vs. chemically stressed pixels based on spectral signatures.	**Heatmap Rendering:** Build the UI components to overlay colored pixel grids onto the 3D map based on mock backend JSON responses.
Mid-Project Review	**Extraction Audit:** Prove the pipeline successfully handles the massive memory requirements of 3D tensor batches during training.	**GIS Validation:** Ensure the Deck.gl overlay aligns perfectly with the real-world GPS coordinates of the farm.
Week 3	**Vision Transformer (ViT):** Integrate a self-attention mechanism (ViT) into the model to improve accuracy by finding complex correlations across different light spectrums.	**Analytics Panel:** Build sidebar charts showing the total acreage at risk and the specific chemical anomalies detected.
Week 4	**API & Tiling:** Wrap the model in FastAPI. Implement a "tiling" algorithm to slice massive satellite images into smaller chunks for inference, stitching the results back together.	**Refine & Polish:** Add timeline sliders to allow the user to see the historical progression of the crop's spectral health.
Final Review	A highly advanced fusion of Geospatial Data Engineering and multi-dimensional Deep Learning.	A professional, visually stunning agricultural intelligence platform.

Project 2 - "EconoCausal": Dynamic Pricing via Double Machine Learning

Domain: Economics & Causal AI

Problem Statement: *Standard ML models predict correlation, not causation. If a company builds a model to predict churn, it might offer a $20 discount to a user with a 90% churn probability. However, that user might have churned anyway, wasting the $20. Traditional ML cannot answer: "Would this specific user have stayed IF AND ONLY IF we gave them the discount?"*

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Use Case: A marketing director uses **EconoCausal** to plan a promotional campaign. Instead of using standard predictive ML, the backend uses **Double Machine Learning (DML)** to calculate the Individual Treatment Effect (ITE). The dashboard doesn't show who is likely to buy; it shows exactly how much a user's probability of buying will *increase* if shown a $10 vs $20 discount. The director uses this to allocate the budget exclusively to "Persuadables" (users who only buy because of the discount), maximizing ROI.

Key Modules:

- **Causal Inference Engine (Python, EconML, DoWhy):** Implements advanced econometric algorithms (Double/Debiased Machine Learning) to strip out confounding variables and isolate true causality.
- **Propensity Score Matching:** Statistical pipelines to balance historical observational data, simulating a randomized A/B test without actually running one.
- **Optimization Solver (SciPy):** An algorithm that takes the causal predictions and a fixed budget (e.g., $5,000) and allocates the optimal discount to each user.
- **Experimentation UI (React & Plotly):** A dashboard visualizing the Causal Uplift curves and budget allocation efficiency.

Week-wise Development Plan:

Week	Causal ML & Optimization (EconML, DoWhy)	Visualization & UI (React, Plotly)
Week 1	**Causal Graphing:** Use Microsoft's DoWhy library to define the causal Directed Acyclic Graph (DAG) for a mock retail dataset (identifying Confounders, Treatments, and Outcomes).	**Dashboard Scaffolding:** Initialize React app. Create views for uploading historical campaign data and setting budget constraints.
Week 2	**Double ML Training:** Train the EconML Double Machine Learning models (using Random Forests or LightGBM as the base estimators) to estimate the Individual Treatment Effect (ITE).	**Uplift Curves:** Integrate Plotly.js to render Qini curves and Uplift charts, visually proving the causal model outperforms a random rollout.
Mid-Project Review	**Causal Audit:** Prove mathematically (via Refutation Tests in DoWhy) that the model has successfully isolated the causal effect from hidden confounders.	**Data Load:** Validate that the dashboard can dynamically filter and chart thousands of customer ITE scores.
Week 3	**Prescriptive Optimization:** Write a SciPy optimization script that maximizes total predicted revenue subject to a strict	**Allocation Matrix:** Build the UI view showing the final "Prescription"—a table

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	marketing budget constraint, assigning personalized discounts.	of users and their mathematically optimal discount amount.
Week 4	**API & Packaging:** Wrap the causal engine in a REST API. Build an automated data drift detector that warns if the underlying customer behavior changes.	**Refine & Polish:** Polish the dashboard, adding simple natural language summaries (e.g., "This strategy saves $4,200 compared to blanket targeting").
Final Review	A masterclass in moving beyond predictive AI into the bleeding edge of Causal and Prescriptive AI.	A highly strategic business intelligence tool ready for enterprise marketing or finance teams.

Project 3 - "QuantFormer": Multimodal Order Book & Sentiment Transformer

Domain: Quantitative Finance (FinTech) & Time-Series Forecasting

Problem Statement: *Predicting short-term stock movements requires analyzing Level 2 (L2) Limit Order Book depth (the exact supply and demand at different price tiers). Traditional time-series models (like ARIMA or LSTMs) suffer from "catastrophic forgetting" on high-frequency data and cannot simultaneously ingest numerical market data alongside unstructured, streaming news text.*

Use Case: A quantitative trader monitors the **QuantFormer** interface. The backend streams raw Nasdaq tick data and live Reuters news headlines. The AI architecture—a Temporal Fusion Transformer (TFT)—ingests the numerical order book depth while simultaneously using FinBERT to encode the sentiment of the breaking news. By fusing these two modalities, the Transformer predicts a sudden liquidity collapse (a "Flash Crash") 5 minutes into the future, automatically halting automated trading strategies to prevent catastrophic losses.

Key Modules:

- **Multimodal Data Ingest (Apache Kafka):** High-throughput ingestion of both numerical L2 market data and textual financial news streams.
- **Temporal Fusion Transformer (PyTorch Forecasting):** A highly advanced Transformer model designed specifically for multi-horizon time-series forecasting, utilizing attention mechanisms to weigh the importance of past market ticks.
- **NLP Encoder (HuggingFace FinBERT):** A specialized LLM that converts breaking news headlines into dense vector embeddings in real-time.
- **Live Trading Dashboard (React, Lightweight Charts):** A professional financial UI displaying candlestick charts, L2 order book depth, and the model's 5-minute confidence intervals.

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Week-wise Development Plan:

Week	Deep Learning & Time-Series (PyTorch, FinBERT)	Financial UI (React, TradingView APIs)
Week 1	**Data Engineering:** Build Kafka producers to stream mock high-frequency L2 Order Book data and a separate stream for mock financial news headlines.	**Trading UI Setup:** Integrate TradingView's Lightweight Charts library into React to render professional, high-performance financial candlesticks.
Week 2	**NLP Integration:** Load HuggingFace FinBERT. Build a pipeline to convert the streaming news text into sentiment embeddings (-1 to 1 scores + dense vectors).	**Order Book DOM:** Build the "Depth of Market" (DOM) component, visualizing the active Bid and Ask volume at various price levels.
Mid-Project Review	**Fusion Audit:** Successfully align the timestamps of the high-frequency tick data with the asynchronous news embeddings into a single PyTorch DataLoader.	**Chart Performance:** Prove the frontend charting library can handle 10+ updates per second without blocking the browser thread.
Week 3	**TFT Architecture:** Implement the Temporal Fusion Transformer using PyTorch Forecasting. Train the model to predict the price 5 steps into the future, utilizing the attention mechanism.	**Predictive Overlays:** Overlay the model's predictions directly onto the candlestick chart, including upper and lower confidence bounds (quantiles).
Week 4	**Attention Interpretability:** Extract the "Attention Weights" from the trained model to understand *why* it made a prediction (e.g., did it react to the order book depth or a specific news headline?).	**Interpretability UI:** Build a sidebar that explains the model's logic (e.g., "Prediction heavily influenced by Negative News at 10:04 AM").
Final Review	An incredibly complex integration of Natural Language Processing and high-frequency Time-Series forecasting.	A slick, institutional-grade quantitative trading dashboard.

Common Features Across All Projects

- **Beyond Standard Tabular Data:** These projects force you to abandon standard CSV datasets (like Titanic or Housing Prices) and work with highly complex structures: 3-Dimensional Hyperspectral Arrays, Causal Treatment matrices, and High-Frequency Multimodal Streams.
- **Cutting-Edge Algorithms:** Utilizing the absolute forefront of machine learning research, including Double Machine Learning (EconML), Vision Transformers (ViT), and Temporal Fusion Transformers (TFT).

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- **Interpretability & Trust:** Advanced AI in the enterprise requires explainability. These projects incorporate features like Causal Refutation Tests and Attention Weight Extraction to prove *why* the AI made a specific decision.
- **GPU-Accelerated Engineering:** Heavy utilization of PyTorch and CUDA to handle massive matrix multiplications and deep learning training loops efficiently.

Overall Project Summary

Project	Target AI/ML Domain	Technology Focus
"TerraSpectra": Crop Disease Forecaster	**Computer Vision & Geospatial**	PyTorch, 3D-CNNs, Vision Transformers (ViT), Rasterio, Deck.gl
"EconoCausal": Dynamic Pricing Engine	**Causal Inference & Economics**	Python, EconML, DoWhy, SciPy Optimization, Plotly
"QuantFormer": Order Book Forecaster	**Time-Series & FinTech**	PyTorch Forecasting (TFT), HuggingFace FinBERT, Kafka, React

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